

ONLINE APPENDIX INFORMATIONAL INERTIA AND THE TAYLOR PRINCIPLE

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1. ESSENTIAL FREQUENCY DOMAIN PROPERTIES OF DISCRETE TIME SERIES

Here we present an (incomplete) introduction, following [Priestly \(1981\)](#), [Ahlfors \(1979\)](#), [Oppenheim, Schafer, and Buck \(1999, ch. 3\)](#), [Oppenheim, Willsky, and Nawab \(1996, ch. 10\)](#), [Hamilton \(1994, ch. 6\)](#), [Sargent \(1987, ch. XI\)](#), and [Shumway and Stoffer \(2011\)](#) to the z-transform and discrete time Fourier transform as it will pertain to our analysis of the determinacy of linear DSGE models. These transforms discern the frequency content and temporal dependencies of a given sequence and, hence, can be used in the analysis of discrete-time series. The autocovariance and autocorrelation functions play the pivotal role in understanding the temporal relationships within a time series and the key element we will introduce here that will be essential for understanding how the sticky information model functions in the frequency domain is the property of scaling in the z-domain.

Our basic assumptions follow, e.g., [Priestly \(1981, ch. 4.11.\)](#) or [Shumway and Stoffer \(2011, Appendix C\)](#), for mean zero, linearly regular covariance stationary stochastic processes with absolutely continuous spectral distribution functions. Let y_t be such a process, then

$$y_t = \int_{-\pi}^{\pi} e^{it\omega} dZ(\omega) \quad (\text{A.1})$$

where $dZ(\omega)$ is a mean zero, random orthogonal increment process with $E[|dZ(\omega)|^2] = h(\omega)d\omega$ and $E[dZ(\omega_1)dZ(\omega_2)^*] = 0$, for $\omega_1 \neq \omega_2$. Assume that the autocovariance function

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is absolutely summable

$$\sum_{m=-\infty}^{\infty} |R_y(m)| < \infty \quad (\text{A.2})$$

where the autocovariance function of a discrete-time series y_t is defined as

$$R_y(m) = \text{Cov}(y_t, y_{t-m}) = E(y_t - \mu_y)(y_{t-m} - \mu_y) \quad (\text{A.3})$$

then the spectral distribution function $Z(\omega)$ is absolutely continuous such that $dZ(\omega) = f_y(\omega)d\omega$ and $f_y(\omega)$ is the spectral density given by

$$f_y(\omega) = \sum_{m=-\infty}^{\infty} R_y(m)e^{-i\omega h}, \quad -\pi \leq \omega \leq \pi \quad (\text{A.4})$$

[Whiteman \(1983\)](#) assumes, and we follow, that solutions for y_t are sought in the space spanned by time-independent square-summable linear combinations of the process(es) fundamental for the driving process, that is H^2 or Hardy space.¹ Let ϵ_t be such a mean zero fundamental process with variance σ_ϵ^2 . Its spectral density is thus

$$f_\epsilon(\omega) = \sum_{m=-\infty}^{\infty} R_\epsilon(m)e^{-i\omega h} = \frac{1}{2\pi} \sigma_\epsilon^2 \quad (\text{A.5})$$

Then an H^2 solution for an endogenous variable, y_t , is of the form

$$y_t = y(L)\epsilon_t = \sum_{j=0}^{\infty} y_j \epsilon_{t-j} \quad (\text{A.6})$$

with $\sum_{j=0}^{\infty} y_j^2 < \infty$ and L the lag operator $Ly_t = y_{t-1}$.² Following, e.g., [Sargent \(1987, ch. XI\)](#) the Riesz-Fischer Theorem gives an equivalence (a one-to-one and onto transformation) between the space of squared summable sequences $\sum_{j=0}^{\infty} y_j^2 < \infty$ and the space of analytic functions in unit disk $y(z)$ corresponding to the z -transform of the sequence, $y(z) = \sum_{j=0}^{\infty} y_j z^j$.

Given a discrete series y_j with samples taken at equally spaced intervals, its z -transform $y(z)$ is defined as

$$y(z) = \sum_{j=0}^{\infty} y_j z^j \quad (\text{A.7})$$

¹See, e.g., [Han, Tan, and Wu \(2022\)](#) for a more formal introduction.

²Note that we are abusing notation somewhat and choosing to use the same letter y to refer to a discrete time series, y_t , as well as that variable's transform function $y(z)$ or MA representation/response to a fundamental process j periods ago, y_j . This serves to save on the verbosity of notation, which might otherwise read $y_t = \sum_{j=0}^{\infty} \delta_j^y \epsilon_{t-j}$ following, e.g., [Meyer-Gohde \(2010\)](#).

where z is a complex variable, and the sum extends from 0 to infinity, following the convention used in [Hamilton \(1994, ch. 6\)](#) and [Sargent \(1987, ch. XI\)](#).³ By evaluating the z -transform on the unit circle in the complex plane ($z = e^{-i\omega}$, where ω is the angular frequency and i the complex number $\sqrt{-1}$), we obtain the discrete-time Fourier transform (DTFT). The DTFT $y(e^{-i\omega})$ is given by

$$y(e^{-i\omega}) = \sum_{j=0}^{\infty} y_j e^{-i\omega j} \quad (\text{A.8})$$

The DTFT reveals the spectral characteristics of the sequence in terms of its frequency components.

The connection between the autocovariance function and the Fourier transformation of the z -transform evaluated on the unit circle ($z = e^{-i\omega}$) can be established by manipulating the equations

$$R_y(m) = \int_{-\pi}^{\pi} f_y(\omega) e^{im\omega} d\omega \quad (\text{A.9})$$

Hence for our mean zero fundamental process ϵ_t

$$R_\epsilon(m) = \int_{-\pi}^{\pi} f_\epsilon(\omega) e^{im\omega} d\omega = \int_{-\pi}^{\pi} \frac{1}{2\pi} \sigma_\epsilon^2 e^{im\omega} d\omega = \frac{1}{2\pi} \sigma_\epsilon^2 \int_{-\pi}^{\pi} e^{im\omega} d\omega = \begin{cases} \sigma_\epsilon^2 & \text{for } m = 0 \\ 0 & \text{otherwise} \end{cases} \quad (\text{A.10})$$

Now return to $y_t = y(L)\epsilon_t = \sum_{j=0}^{\infty} y_j \epsilon_{t-j}$ and recall $y_t = \int_{-\pi}^{\pi} e^{it\omega} dZ_y(\omega)$ and analogously $\epsilon_t = \int_{-\pi}^{\pi} e^{it\omega} dZ_\epsilon(\omega)$ so therefore it must hold that

$$\int_{-\pi}^{\pi} e^{it\omega} dZ_y(\omega) = \int_{-\pi}^{\pi} y(e^{-i\omega}) e^{it\omega} dZ_\epsilon(\omega) \Rightarrow dZ_y(\omega) = y(e^{-i\omega}) dZ_\epsilon(\omega) \quad (\text{A.11})$$

Multiplying both sides by their complex conjugates and taking expectations gives

$$E[dZ_y(\omega) dZ_y(\omega)^*] = E[y(e^{-i\omega}) y(e^{-i\omega})^* dZ_\epsilon(\omega) dZ_\epsilon(\omega)^*] \quad (\text{A.12})$$

$$f_y(\omega) = |y(e^{-i\omega})|^2 f_\epsilon(\omega) = |y(e^{-i\omega})|^2 \frac{1}{2\pi} \sigma_\epsilon^2 \quad (\text{A.13})$$

We can insert this directly into (A.9) above to yield

$$R_y(m) = \sigma_\epsilon^2 \frac{1}{2\pi} \int_{-\pi}^{\pi} |y(e^{-i\omega})|^2 e^{im\omega} d\omega \quad (\text{A.14})$$

³The discrete signal processing and systems theory literature works in negative exponents of z , see [Oppenheim, Schaffer, and Buck \(1999, ch. 3\)](#) and [Oppenheim, Willsky, and Nawab \(1996, ch. 10\)](#). [Al-Sadoon \(2024\)](#) follows this convention and interprets the operator being applied as the forward operator. We maintain the more familiar approach in working with the lag operator which results in our use of positive exponents in z .

where $y(e^{-i\omega})$ and $y^*(e^{i\omega})$ denote the DTFT of y_j and its complex conjugate, respectively. This relationship allows us to analyze the temporal dependencies in a time series.

To see the content of the spectral representation - and in particular, how scaling in the z domain that we use in the main text for the spectral representation of [Mankiw and Reis's \(2002\)](#) sticky information in (29) affects a series' autocovariance - we will examine an AR(1) example⁴

$$y_t = \rho y_{t-1} + \epsilon_t \quad (\text{A.15})$$

where y_t is the current value of the process, y_{t-1} is the previous value, ρ is the autoregressive parameter assumed less than one in absolute value, and ϵ_t is the white noise error term at time t with standard deviation σ_ϵ . The infinite MA representation is given by

$$y_t = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-j} = \left(\sum_{j=0}^{\infty} \rho^j L^j \right) \epsilon_t \quad (\text{A.16})$$

where L is again the lag operator ($L\epsilon_t = \epsilon_{t-1}$). This gives us (A.8) with $y_j = \rho^j$ and $L = z$ an operator defined on the unit circle.

We can use the z -transform and Fourier transformation to calculate the autocovariance of our AR(1) process. Taking the z -transform of both sides of (A.15), we have

$$y(z) = \rho z y(z) + 1 \Rightarrow y(z) = \frac{1}{1 - \rho z} \quad (\text{A.17})$$

where $y(z)$ is the z -transform of the AR(1) transfer function. Now, we can calculate the autocovariance using the square of the absolute value of the Fourier transform of the transfer function as in (A.14). Accordingly, $R_y(m)$ can be expressed as

$$R_y(m) = \frac{\sigma_\epsilon^2}{2\pi} \int_{-\pi}^{\pi} \left| y(e^{-i\omega}) \right|^2 e^{im\omega} d\omega = \frac{\sigma_\epsilon^2}{2\pi} \int_{-\pi}^{\pi} \left| \frac{1}{1 - \rho e^{-i\omega}} \right|^2 e^{im\omega} d\omega \quad (\text{A.18})$$

which can be written as a contour integral along the unit circle parameterized by $\zeta = e^{i\omega}$

$$R_y(m) = \frac{\sigma_\epsilon^2}{2\pi i} \oint_{|\zeta|=1} \frac{\zeta^{m-1}}{(1 - \rho\zeta^{-1})(1 - \rho\zeta)} d\zeta = \frac{\sigma_\epsilon^2}{2\pi i} \oint_{|\zeta|=1} \frac{\zeta^m}{(\zeta - \rho)(1 - \rho\zeta)} d\zeta \quad (\text{A.19})$$

which can be evaluated by residues⁵ for $m \neq 0$. The function $\zeta^{m-1}/|1 - \rho\zeta^{-1}|^2$ has a simple pole inside the contour (unit circle) at $\zeta = \rho$. The residue at $\zeta = \rho$ is:

$$\text{Res}_{\zeta=\rho} \left[\zeta^{m-1}/|1 - \rho\zeta^{-1}|^2 \right] = \text{Res}_{\zeta=\rho} \left[\zeta^m / ((\zeta - \rho)(1 - \rho\zeta)) \right] = \rho^m / (1 - \rho^2) \quad (\text{A.20})$$

⁴An additional ARMA(2,2) example can be found later in this appendix.

⁵The residue of a function $f(\zeta)$ at a pole ζ_0 of order k is given by $\text{Res}_{\zeta=\zeta_0}[f(\zeta)] = \frac{1}{(k-1)!} \lim_{\zeta \rightarrow \zeta_0} \frac{d^{k-1}}{d\zeta^{k-1}} ((\zeta - \zeta_0)^k f(\zeta))$ and the contour integral along γ is $\frac{1}{2\pi i} \oint_\gamma f(\zeta) d\zeta = \sum_j \text{Res}_{\zeta=\zeta_j}[f(\zeta)]$ where the sum is over all the singularities - ζ_j - enclosed by γ , see [Ahlfors \(1979, ch. 4.5\)](#).

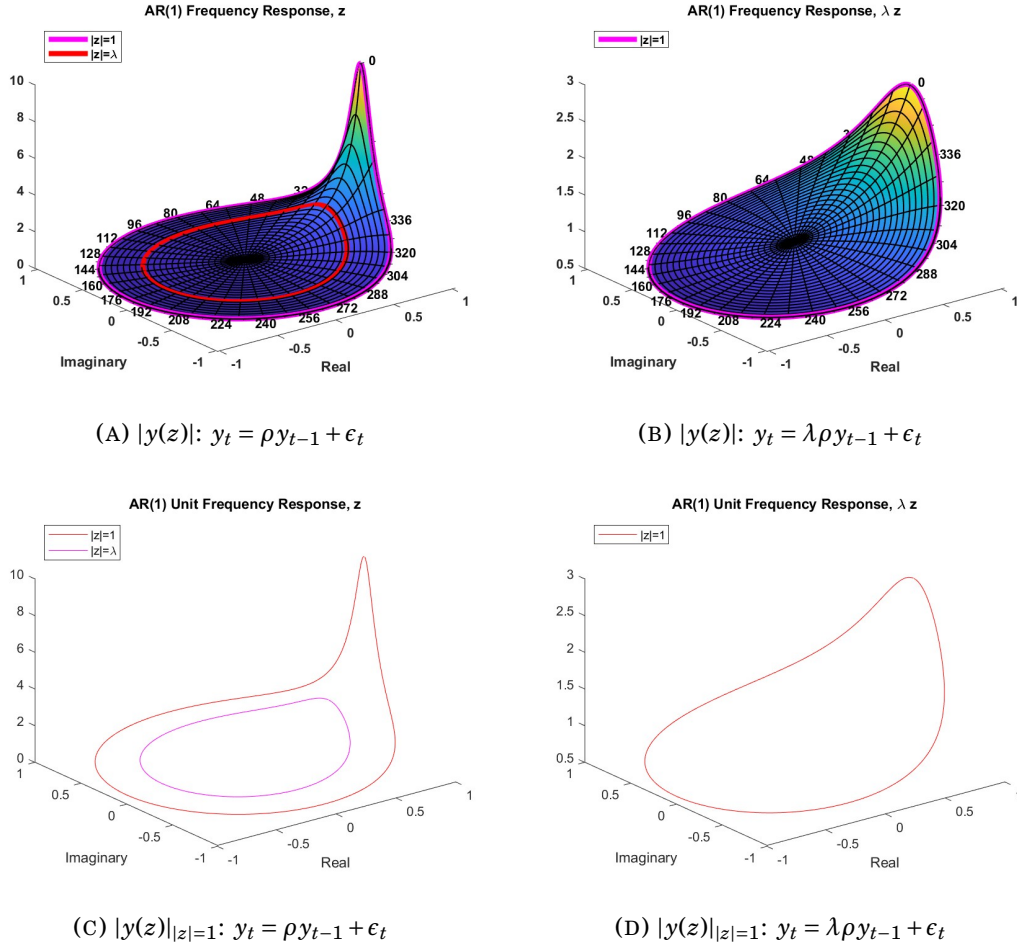


FIGURE 1. AR(1) - Transfer Functions on the Unit Disk

The values $\rho = 0.9$ and $\lambda = 0.7$ were used

which gives the autocovariance function of y_t as

$$R_y(m) = \sigma^2 \times \text{Res}_{\zeta=\rho} = \sigma^2 \rho^m / (1 - \rho^2) \quad (\text{A.21})$$

The same value we would obtain using time domain methods.

Figure 1 plots the (absolute value of the) transfer function $|y(z)|$, $|z| \leq 1$ for two values of ρ . In figure 1a, the absolute value of the transfer function is plotted with $\rho = 0.9$ and in figure 1b with the autoregressive parameter dampened by $\lambda = 0.7$. The values on the unit circle can be found in the lower two panels, figures 1c and 1d, which can be used in (A.18) to determine the autocovariances.

Among the properties of the z-transform - see, e.g., [Oppenheim, Schafer, and Buck \(1999, ch. 3\)](#) and [Oppenheim, Willsky, and Nawab \(1996, ch. 10\)](#), the one that will be both particularly relevant for interpreting sticky information in the next section (and is less known to economists) is that of scaling in the z domain. Proposition 1 tells us that multiplying a sequence with a given region of convergence and set of poles and zeros by

an exponential sequence in λ scales the region of convergence and the poles and zeros of y by λ .

Proposition 1 (Scaling in the z domain). *Given a z -transform of a sequence with a region of convergence R*

$$y(z) = \sum_{j=0}^{\infty} y_j z^j \quad (\text{A.22})$$

the scaled sequence

$$y(\lambda z) = \sum_{j=0}^{\infty} y_j \lambda^j z^j \quad (\text{A.23})$$

has a region of convergence $R/|\lambda|$ and if $y(z)$ has a pole (or zero) at a , then $y(\lambda z)$ has a pole (or zero) at λa .

Proof. See [Oppenheim, Willsky, and Nawab \(1996, p. 768\)](#) and note the difference in convention with the signal processing literature developing series in the inverse of z in contrast to the time series literature - e.g., [Sargent \(1987, ch.XI\)](#) and [Hamilton \(1994, ch. 6\)](#). $\square$

To understand the effects of scaling in the frequency domain, consider the following example. Let A_t be a mean zero, linearly regular covariance stationary stochastic process with known Wold representation given by

$$A_t = A(L)\epsilon_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i} = \sum_{i=0}^{\infty} a_i L^i \epsilon_t \quad (\text{A.24})$$

Compare this with the process

$$B_t = A(\lambda L)\epsilon_t = \sum_{i=0}^{\infty} a_i \lambda^i \epsilon_{t-i} = \sum_{i=0}^{\infty} a_i (\lambda L)^i \epsilon_t \quad (\text{A.25})$$

The autocovariance of A_t is given by

$$c_A(h) = \sum_{i=-\infty}^{\infty} a_i a_{i+h} \sigma_{\epsilon}^2 \quad (\text{A.26})$$

and of B_t by

$$c_B(h) = \sum_{i=-\infty}^{\infty} \lambda^i a_i \lambda^{i+h} a_{i+h} \sigma_{\epsilon}^2 = \lambda^h \sum_{i=-\infty}^{\infty} \lambda^{2i} a_i a_{i+h} \sigma_{\epsilon}^2 \quad (\text{A.27})$$

Inspection shows that for $0 < \lambda < 1$, $c_B(h) < c_A(h)$ and that $c_B(h)$ is decreasing in h at a rate λ .

This is directly exemplified by the AR(1) process above. Figure 1 plots $|y(z)|$ for $y_t = \rho y_{t-1} + \epsilon_t$ in the left panels and $|y(\lambda z)|$ on the right. Notice that the entire transfer function inside the closed unit disk for $y_t = \lambda \rho y_{t-1} + \epsilon_t$ can be found as the transfer function of

$y_t = \rho y_{t-1} + \epsilon_t$ inside the circle with radius λ . That is, λ scales the transfer function and in this case with $|\lambda| < 1$ towards the origin - that is, away from the unconditional response $|y(1)|$ to shocks at all time horizons and towards the impact response $|y(0)|$ of the process to contemporaneous shocks.

The final and, for our determinacy analysis later, crucial property to observe is that this dampening is not bidirectional. If $|y(z)|$ is well defined (analytic) on the unit disk, so too will $|H(\lambda z)|$ be for $|\lambda| < 1$. Defining $\tilde{z} = \lambda z$, $|y(\tilde{z})|$ being well defined (analytic) on the unit disk does not allow us conclude the same about $|y(\frac{1}{\lambda}\tilde{z})|$ for $|\lambda| < 1$, as $\frac{1}{\lambda}\tilde{z}$ goes past the unit circle. That is, following Proposition 1, λ scales the region of convergence and if the process defined by $y(z)$ has a region of convergence from the origin out to the unit circle, then the process associated with $H(\frac{1}{\lambda}z)$ has a region of convergence out only to $|\lambda| < 1$.

2. FREQUENCY DOMAIN SOLUTION OF FORWARD-LOOKING MODELS

Having laid out the basic properties and paid specific attention to the scaling in the z domain property, we now turn to solving rational expectations models in the frequency domain following Whiteman (1983) - see also Taylor (1986, ch. 2.3) for an approachable introduction with direct comparisons to other methods.

Starting with expectations, the Wiener-Kolmogorov prediction formula gives us $E_t[y_{t+n}] = E_t\left[\sum_{j=0}^{\infty} y_j \epsilon_{t-j+n}\right] = \sum_{j=0}^{\infty} y_{j+n} \epsilon_{t-j}$. The Wiener-Kolmogorov prediction formula of “plussing” gives the frequency domain version

$$\mathcal{Z}\{E_t[x_{t+1}]\} = \left[\frac{x(z)}{z} \right]_+ = \frac{1}{z}(x(z) - x(0)) \quad (\text{B.28})$$

where $_+$ is the annihilation operator, see Sargent (1987) and Hamilton (1994).

Consider a backward and forward looking model in y_t and ϵ_t

$$aE_t y_{t+1} + b y_t + c y_{t-1} + \epsilon_t = 0 \quad (\text{B.29})$$

The same process is presented in the z domain as

$$a \frac{1}{z}(y(z) - y_0) + b y(z) + c z y(z) + 1 = 0 \quad (\text{B.30})$$

Rearranging allows us to reduce the solution to this model as

$$a(y(z) - y_0) + b z y(z) + c z^2 y(z) + z = 0 \Leftrightarrow (a + b z + c z^2) y(z) = a y_0 - z \quad (\text{B.31})$$

$$(a - a(\lambda_1 + \lambda_2)z + a\lambda_1\lambda_2 z^2) y(z) = a y_0 - z \Leftrightarrow (1 - \lambda_1 z)(1 - \lambda_2 z) y(z) = y_0 - \frac{z}{a} \quad (\text{B.32})$$

with the initial condition on y_0 to be determined.

We will require that $y(z)$ be analytic inside the unit disk to give us a stationary process y_t causal in ϵ_t . Consider now the following possibilities. If $|\lambda_1|, |\lambda_2| < 1$, then there is no singularity in $y(z)$ inside the unit circle that can be removed to pin down y_0 and, we find that $(1 - \lambda_1 L)(1 - \lambda_2 L)y_t = (y_0 - \frac{L}{a})\epsilon_t$ is necessarily unstable as at most one of the two unstable autoregressive factors $(1 - \lambda_k L)$ could be removed by a particular choice of y_0 - that is, we have non existence of a stable solution. If, however, $|\lambda_1|, |\lambda_2| > 1$, there are two singularities in $y(z)$ inside the unit circle and y_0 cannot be uniquely determined so there are multiple stable solutions - that is, we have indeterminacy. If however, $|\lambda_2| < 1 < |\lambda_1|$, there is one singularity in $y(z)$ inside the unit circle, namely at $z = 1/\lambda_1$, and using the residue theorem⁶ it can be removed to ensure the analyticity of $y(z)$ over the unit disk by setting the boundary condition on y_0 as

$$\lim_{z \rightarrow \frac{1}{\lambda_1}} (1 - \lambda_1 z)(1 - \lambda_2 z)y(z) \stackrel{!}{=} 0 = y_0 - \frac{1}{\lambda_1 a} \Rightarrow y_0 = \frac{1}{\lambda_1 a} \quad (\text{B.33})$$

which determines the unique stable solution for the process on $y(z)$ as

$$y(z) = \frac{1}{1 - \lambda_1 z} \frac{1}{1 - \lambda_2 z} \frac{1}{a} \left(\frac{1}{\lambda_1} - z \right) = \frac{1}{1 - \lambda_2 z} \frac{1}{\lambda_1 a} = \frac{1}{\lambda_1 a} \frac{1}{1 - \lambda_2 z} \quad (\text{B.34})$$

Substituting the lag operator for z to express in the time domain gives us

$$y_t = \frac{1}{\lambda_1 a} \frac{1}{1 - \lambda_2 L} \epsilon_t \Rightarrow y_t = \lambda_2 y_{t-1} + \frac{1}{\lambda_1 a} \epsilon_t \quad (\text{B.35})$$

Hence our requirement that one root be inside and one outside the unit circle gives us the famed [Blanchard and Kahn \(1980\)](#) condition. Underlining the point that deriving the condition in either time or frequency domain neither alters the model itself or the associated conditions for determinacy, but simply allows us to determine unique solutions and boundary conditions of models with different tools.

3. DETERMINACY OF THE STICKY PRICE MODEL IN THE FREQUENCY DOMAIN

First, we close the model with (25), the standard sticky-price Phillips curve,

$$\pi(z) = \beta \frac{1}{z} (\pi(z) - \pi_0) + \kappa y(z) \quad (\text{C.36})$$

and the two foregoing can be summarized in a matrix system as

$$\begin{bmatrix} -\beta & 0 \\ \sigma & 1 \end{bmatrix} \begin{bmatrix} \pi(z) \\ y(z) \end{bmatrix} = \begin{bmatrix} -1 & \kappa \\ \sigma \phi_\pi & 1 + \sigma \phi_y \end{bmatrix} z \begin{bmatrix} \pi(z) \\ y(z) \end{bmatrix} + \begin{bmatrix} -\beta & 0 \\ \sigma & 1 \end{bmatrix} \begin{bmatrix} \pi_0 \\ y_0 \end{bmatrix} \quad (\text{C.37})$$

⁶See [Ahlfors \(1979, ch. 4\)](#).

or equivalently,

$$(I_2 - zA) \begin{bmatrix} \pi(z) \\ y(z) \end{bmatrix} = \begin{bmatrix} \pi_0 \\ y_0 \end{bmatrix} \quad (\text{C.38})$$

where $A = \begin{bmatrix} \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ \sigma(\phi_\pi - \frac{1}{\beta}) & 1 + \frac{\sigma}{\beta}\kappa + \sigma\phi_y \end{bmatrix}$ is the matrix of coefficients. We summarize the condition for determinacy in the following.

Theorem 1 (Sticky Price Determinacy in the Frequency Domain). *The sticky price model, given by (2), (25), with the Taylor rule (1), has a unique, stable equilibrium if and only if*

$$\phi_\pi > 1 - \frac{1 - \beta}{\kappa} \phi_y \quad (\text{C.39})$$

Proof. See the following (cf. time domain result 2) □

To solve the system of equations in (C.38) we first decompose the matrix A and then use Cauchy's residue theorem as above to determine π_0 and y_0 , the initial conditions for inflation and the output gap. Define $\rho_i = \text{eig}(A)$. Iff ρ_i , $i = 1, 2$ there are two removable singularities. Decompose matrix A into its eigenvalues, and its eigenvector-matrix V as

$$A = V \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} V^{-1} = V \Lambda V^{-1} \quad (\text{C.40})$$

Similar to Klein (2000) we define

$$\begin{bmatrix} w(z) \\ u(z) \end{bmatrix} = V^{-1} \begin{bmatrix} \pi(z) \\ y(z) \end{bmatrix} \quad \text{for } z = 0, 1, 2, \dots \quad (\text{C.41})$$

Substituting into our equation system, (C.38) gives

$$(I_2 - zV\Lambda V^{-1})V \begin{bmatrix} w(z) \\ u(z) \end{bmatrix} = V \begin{bmatrix} w_0 \\ u_0 \end{bmatrix} \quad (\text{C.42})$$

which can be rewritten and redefined as

$$(I_2 - z\Lambda) \begin{bmatrix} w(z) \\ u(z) \end{bmatrix} = \begin{bmatrix} w_0 \\ u_0 \end{bmatrix} \quad (\text{C.43})$$

The diagonality of the foregoing yields two independent equations

$$(1 - z\rho_1)w(z) = w_0 \text{ and } (1 - z\rho_2)u(z) = u_0 \quad (\text{C.44})$$

If both eigenvalues, $|\lambda_1|$ and $|\lambda_2| > 1$, we can eliminate the singularities via

$$\lim_{z \rightarrow 1/\lambda_1} (1 - z\lambda_1)w(z) = 0 \text{ and } \lim_{z \rightarrow 1/\lambda_2} (1 - z\lambda_2)u(z) = 0 \quad (\text{C.45})$$

pinning down the two conditions $w_0 = 0$ and $u_0 = 0$. From our definition (C.41) and equation (C.43) we can therefore deduce

$$\begin{bmatrix} \pi_0 \\ y_0 \end{bmatrix} = V \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (\text{C.46})$$

uniquely defining $\pi_0 = 0$ and $y_0 = 0$.

The Schur-Cohn criteria can be applied to ascertain whether both eigenvalues, λ_1 and λ_2 , indeed do lie outside the unit circle (see LaSalle, 1986, p.28). These criteria, expressed in terms of A are $|\det(A)| > 1$ and $|\text{tr}(A)| < 1 + \det(A)$. As

$$\det(A) = \frac{1}{\beta}(1 + \sigma\phi_y + \kappa\sigma\phi_\pi) > 1 \text{ and } \text{tr}(A) = \frac{1}{\beta} + \frac{\sigma\kappa}{\beta} + 1 + \sigma\phi_y > 1 \quad (\text{C.47})$$

The condition $|\det(A)| > 1$ necessarily holds and $|\text{tr}(A)| < 1 + \det(A)$ holds if

$$1 < \frac{1 - \beta}{\kappa}\phi_y + \phi_\pi. \quad (\text{C.48})$$

Hence, determinacy in the sticky price model demands

$$1 - \frac{1 - \beta}{\kappa}\phi_y < \phi_\pi. \quad (\text{C.49})$$

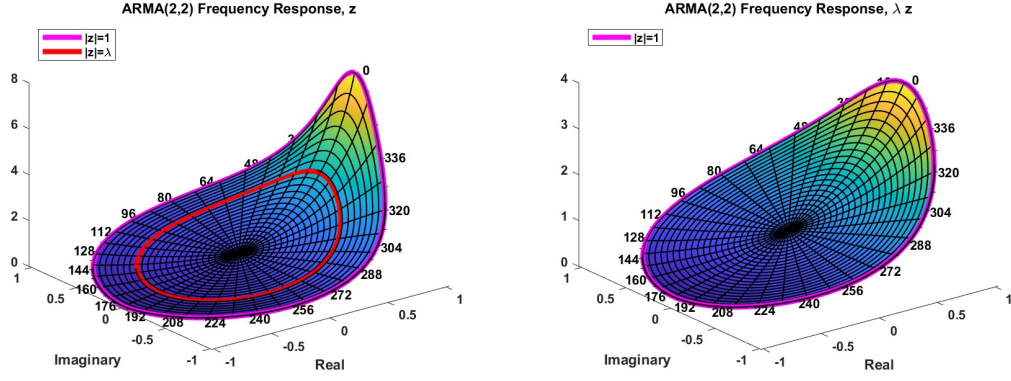
Given the Taylor rule (1), the monetary authority can target inflation as well as the output gap to stabilize the economy - Woodford (2003, pp. 254–255), “... indeed, a large enough [response to] *either* [the output gap or inflation] suffices to guarantee determinacy”. Indeed, the real rate can be raised in response to an off equilibrium inflation increase even by responding to output movements alone. Notice that this possibility disappears if $\beta = 1$ - however this is misleading as although an *average* long-run tradeoff disappears in this case, a dynamic one remains $\frac{\pi_t - E_t \pi_{t+1}}{\kappa} = y_t$ which monetary policy needs for its targeting of inflation (or output) at different horizons to translate into a response to current inflation as we see in our analysis of extended Taylor rules in the main text.

4. AR(2) EXAMPLE OF SCALING IN THE Z DOMAIN

While one might be tempted to dismiss the AR(1) result as a coincidence of the exponential scaling inherently involved with an AR(1) process, examination of a more complicated process, such as an ARMA(2,2) ought to dissuade this temptation

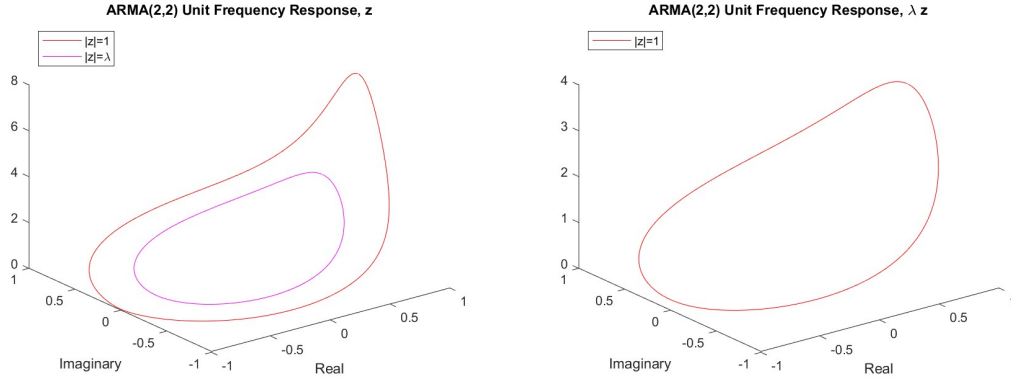
$$y_t + \rho_1 y_{t-1} + \rho_2 y_{t-2} = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} \quad (\text{D.50})$$

Figure 2 contains the same four panels as for the AR(1) above and, again, the dampening property of $|\lambda| < 1$ is displayed. The transfer function of the ARMA(2,2) is scaled towards



$$(A) |y(z)|: (1 + \rho_1 L + \rho_2 L^2)y_t = (1 + \theta_1 L + \theta_2 L^2)\epsilon_t$$

$$(B) |y(z)|: (1 + \lambda \rho_1 L + \lambda^2 \rho_2 L^2)y_t = (1 + \lambda \theta_1 L + \lambda^2 \theta_2 L^2)\epsilon_t$$



$$(C) |y(z)|_{|z|=1}: (1 + \rho_1 L + \rho_2 L^2)y_t = (1 + \theta_1 L + \theta_2 L^2)\epsilon_t$$

$$(D) |y(z)|_{|z|=1}: (1 + \lambda \rho_1 L + \lambda^2 \rho_2 L^2)y_t = (1 + \lambda \theta_1 L + \lambda^2 \theta_2 L^2)\epsilon_t$$

FIGURE 2. ARMA(2,2) - Transfer Functions on the Unit Disk

The values $\rho_1 = 1.1$, $\rho_2 = -0.28$, $\theta_1 = 0.6$, $\theta_2 = -0.25$, and $\lambda = 0.7$ were used

the origin by $|\lambda| < 1$. Comparing the upper two panels, the scaling of the z axis instantly reveals the dampening of the associated ARMA(2,2) on the right with L replaced by λL and by noticing that this transfer function is a subset of the original ARMA(2,2) transfer function, out only to $|\lambda|$ instead of 1.

The final, and for our determinacy analysis crucial, property to observe is that this dampening is not bidirectional. If $|y(z)|$ is well defined (analytic) on the unit disk, so too will $|H(\lambda z)|$ be for $|\lambda| < 1$. Defining $\tilde{z} = \lambda z$, $|y(\tilde{z})|$ being well defined (analytic) on the unit disk does not allow us to conclude the same about $|y(\frac{1}{\lambda}\tilde{z})|$ for $|\lambda| < 1$, as $\frac{1}{\lambda}\tilde{z}$ goes past the unit circle. That is, following Proposition 1, λ scales the region of convergence and if the process defined by $y(z)$ has a region of convergence from the origin out to the unit circle, then the process associated with $H(\frac{1}{\lambda}z)$ has a region of convergence out only to $|\lambda| < 1$.

A. Additional examples of determinacy in the frequency domain

We briefly demonstrate the requirement of analyticity of the z-transform in the frequency domain in relation to known requirements in the time domain in order to establish intuition. Consider first an autoregressive process of order 1, an AR(1) process:

$$y_t = \rho y_{t-1} + \epsilon_t, \quad \epsilon_t \sim WN(0, \sigma^2) \quad (\text{D.51})$$

which can be rewritten as

$$y_t = \sum_{j=0}^{\infty} L^j y_j \epsilon_t. \quad (\text{D.52})$$

The AR(1) process in the frequency domain, see above, is given by applying the z-transform:

$$y(z) = \rho y(z) + 1 \quad (\text{D.53})$$

$$y(z) = \frac{1}{1 - \rho z} \quad (\text{D.54})$$

$y(z)$ analytic inside the unit disk if $|\rho| < 1$ and determines the solution to the autoregressive process.

Now consider a forward-looking process:

$$y_t = \alpha E_t y_{t+1} + \epsilon_t \quad (\text{D.55})$$

whereby the forecast can be rewritten in terms of deviations from the driving process:

$$E_t y_{t+1} = y_{t+1} - y_0 \epsilon_{t+1} = \frac{1}{L} \left(\sum_{j=0}^{\infty} L^j y_j - y_0 \right) \epsilon_t. \quad (\text{D.56})$$

In the frequency domain the forward-looking process is described by:

$$y(z) = \alpha \frac{1}{z} (y(z) - y_0) + 1 \quad (\text{D.57})$$

where $y_0 = y(0)$ is the value of the y at frequency 0 and simultaneously presents the initial condition of the stationary process. To determine a solution we solve for $y(z)$:

$$\left(1 - \frac{1}{\alpha} z \right) y(z) = y_0 - \frac{z}{\alpha} \quad (\text{D.58})$$

$$y(z) = \left(1 - \frac{1}{\alpha} z \right)^{-1} \left(y_0 - \frac{z}{\alpha} \right) \quad (\text{D.59})$$

whereby y_0 is not determined yet. If $|\alpha| < 1$, then for $z = \alpha$ there is a removable singularity inside the unit disk and we can solve for a boundary condition on y_0 :

$$\lim_{z \rightarrow \alpha} \left(1 - \frac{1}{\alpha} z \right) y(z) = 0 \quad (\text{D.60})$$

giving rise to the initial condition of $y_0 = 1$. The solution to our process in the frequency domain is then determined as:

$$y(z) = \frac{1 - \frac{z}{\alpha}}{1 - \frac{z}{\alpha}} = 1. \quad (\text{D.61})$$

In the time domain the equivalent unique stationary solution is given by:

$$y_t = \epsilon_t. \quad (\text{D.62})$$

Compare with [Blanchard \(1979\)](#).

5. ALTERNATIVE DERIVATION OF FREQUENCY DOMAIN STICKY INFORMATION

We can also derive a recursive representation of the lagged expectations of the endogenous variables in (26) as

$$(1 - \lambda) \sum_{i=0}^{\infty} \lambda^i E_{t-i-1}[x_t], \quad x_t = \left(\sum_{j=0}^{\infty} x_j z^j \right) \epsilon_t \quad (\text{E.63})$$

$$= (1 - \lambda) (E_{t-1}[x_t] + \lambda E_{t-2}[x_t] + \lambda^2 E_{t-3}[x_t] + \dots) \quad (\text{E.64})$$

Applying the Wiener-Kolmogorov prediction formula to the lagged expectations (27), equation (E.64) gives the frequency domain representation as:

$$(1 - \lambda) (x(z) - x_0 + \lambda(x(z) - x_0 - zx_1) + \lambda^2(x(z) - x_0 - zx_1 - z^2x_2) + \dots) \quad (\text{E.65})$$

$$= (1 - \lambda) (x(z) + \lambda x(z) + \lambda^2 x(z) + \dots - x_0 - \lambda x_0 - \lambda^2 x_0 \dots - \lambda z x_1 - \lambda^2 z x_1 \dots - \lambda^2 z x_2 \dots)$$

$$= (1 - \lambda) ((1 + \lambda + \lambda^2 + \dots)x(z) - (1 + \lambda + \lambda^2 + \dots)x_0 - \lambda z(1 + \lambda + \lambda^2 + \dots)x_1 - \lambda^2 z^2(1 + \lambda + \lambda^2 + \dots)x_2 - \dots) \quad (\text{E.66})$$

$$= (1 - \lambda) \left(\frac{1}{1 - \lambda} x(z) - \frac{1}{1 - \lambda} x_0 - \frac{\lambda z}{1 - \lambda} x_1 - \frac{\lambda^2 z^2}{1 - \lambda} x_2 - \dots \right) \quad (\text{E.67})$$

$$= x(z) - \sum_{j=0}^{\infty} \lambda^j z^j x_j = x(z) - x(\lambda z) \quad (\text{E.68})$$

Hence, the lagged expectations in (E.64) can be transformed from the time into the frequency domain as:

$$\begin{aligned} (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j E_{t-j-1}[x_{t-1}] &= (1 - \lambda) \left(\frac{z}{1 - \lambda} x(z) - \frac{\lambda z}{1 - \lambda} x_0 - \frac{(\lambda z)^2}{1 - \lambda} x_1 - \dots \right) \\ &= zx(z) - \lambda zx(\lambda z) \end{aligned} \quad (\text{E.69})$$

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